



OPEN Spatio-temporal dynamics of root traits regulate nutrient acquisition and yield of pigeonpea-wheat based system under organic and natural farming

Seema Hodkashia¹, Sarvendra Kumar^{1✉}, Dibakar Mahanta², Tapan Jyoti Purakayastha¹, Rakesh Pandey³, Debashis Mandal¹, Rosin K. G⁴, Anil Kumar Mishra⁴, Sangeeta Paul⁵, Arijit Chowdhuri¹, V. Mubashir Sadiq¹, Sumitra Kumawat¹ & Arun Kumar^{6✉}

Root system development strongly influences nutrient acquisition, yield formation, and these processes are largely influenced by crop management strategies. Meagre information is available on functional traits of roots that regulate nutrient acquisition and productivity of pigeonpea under organic and natural farming systems in semi-arid regions. This study aimed to quantify the spatial and temporal dynamics of root morphological traits, nutrient acquisition and grain yield of pigeonpea under specific crop management strategies. A field experiment was initiated during 2021-22 under a pigeonpea-wheat system with seven treatments viz.; conventional farming (CF), natural farming (NF) and five compost rates (0, 6, 12, 18, and 24 t ha⁻¹). Root samples were collected at 45, 90, and 135 days after sowing (DAS) from two soil depths (0–15 and 15–30 cm) to assess the impact of different treatments on root functional traits, nutrient acquisition, and seed yield of pigeonpea in the year, 2024. The root growth and nutrient acquisition peaked at the flowering stage (90 DAS), subsequently declined during the pod filling stage (135 DAS). Compared with natural farming and conventional farming, compost (12 t ha⁻¹) increased root length density by 23.3 & 8.0%, root surface area by 23.0 & 17.6%, root volume by 22.7 & 20.9%, root fresh weight (RFW) by 22.7 & 15.0%, respectively, in the 0–15 cm soil layer at the flowering stage. The higher nutrient content (phosphorus and potassium) was noticed with the same treatment (0.35 and 0.98%, respectively), followed by compost (6 t ha⁻¹). The order of treatments in terms of nutrient contents were compost (12 t ha⁻¹) > compost (6 t ha⁻¹) > CF > compost (18 t ha⁻¹) > NF > compost (24 t ha⁻¹); in contrast, higher nitrogen content was observed in NF (3.96%) followed by compost (12 t ha⁻¹). Root morphological traits positively correlated ($p < 0.001$) with nutrient uptake, particularly for phosphorus and potassium, suggesting that improved root development enhanced nutrient acquisition from the soil and played a key role in yield formation. Seed yield ranged from 10.1 to 13.4 q ha⁻¹ and crop supplied with compost @ 12 t ha⁻¹ produced 16.5 and 19.2% higher yield than CF and NF, respectively. Regression analysis showed a linear relationship between grain yield and nutrients uptake across all crop growth stages. The seed yield was positively and strongly correlated to P uptake at 45 DAS ($R^2 = 0.85$, $p < 0.001$), followed by the flowering stage ($R^2 = 0.83$) and pod filling stage ($R^2 = 0.75$). Spatio-temporal variation in root traits strongly regulated nutrient acquisition and yield formation in pigeonpea. Optimized compost application (12 t ha⁻¹) enhanced root development and nutrient uptake, with root phosphorus accumulation emerged as the key driver of grain yield, underscoring its importance for sustainable pigeonpea production.

Keywords Conventional farming, Natural farming, Organic farming, Root morphological traits, Sustainable agriculture

¹Division of Soil Science and Agricultural Chemistry, ICAR-Indian Agricultural Research Institute, New Delhi, India. ²Division of Agronomy, ICAR-Indian Agricultural Research Institute, New Delhi, India. ³Division of Plant Physiology, ICAR-Indian Agricultural Research Institute, New Delhi, India. ⁴Water Technology Centre, ICAR-Indian Agricultural Research Institute, New Delhi, India. ⁵Division of Microbiology, ICAR-Indian Agricultural Research Institute, New Delhi, India. ⁶Department of Biosciences, School of Physical & Biological Sciences, Faculty of Science, Technology & Architecture, Manipal University Jaipur, Jaipur, Rajasthan, India. ✉email: sarv.grandee@gmail.com; arun.kumar@jaipur.manipal.edu

Pigeonpea (*Cajanus cajan* L.) is a deep-rooted legume crop and a prime source of protein for populations across Asia, Africa, and America¹. It is the most important legume crop cultivated across the length and breadth of the globe, following common bean, chickpea, field peas, cowpea and lentils². Pigeonpea crop is cultivated on approximately 6.1 million hectares (Mha) worldwide with an annual production of 5 million tons (MT). India is a top producer of pigeonpea, followed by Myanmar, Malawi and Kenya³. India accounts for 72% of the area grown to pigeonpea⁴. It is cultivated on residual soil moisture due to stress tolerance and hardiness, ensuring sustainable yields with minimum inputs. The root growth behaviour is influenced by various inputs like water and nutrients⁵. In current years, research has progressively focused on role of underground root systems in nutrient acquisition within crop science. However, a thorough understanding of effects of fertilizer management practices on root behaviour is very crucial. The improved root traits, including root length, root surface area, root volume and biomass, enhance fertilizer and nutrient use efficiency, thereby influencing nutrient acquisition and ultimately seed yield^{6,7}.

Increased food production in line with the ever-increasing demographic pressure is the need of the hour. Hence, improving crop production is a challenging task in conventional farming systems⁸. Though conventional farming systems (CF) enhance grain yield, their heavy and continued reliance on chemical fertilizers may be ineffective for sustaining long-term soil health by reducing nutrient use efficiency and leading to environmental pollution^{9–11}. These challenges have driven the movement towards eco – farming practices, which aims to achieve soil and plant health while reducing dependence on inorganic inputs. An important part of present study was identification of nutrient sources that can be sustainable and reliable for crop growth and yield. It has been recognised that organic amendments, such as cover crops, compost and natural farming amendments, contain significant amounts of nutrients that could be used to sustain and enhance crop growth¹². In this context, the practice of compost application¹³ and natural farming¹⁴ has been strongly promoted. However, nutrient availability from organic amendments is often variable and less predictable than that from inorganic fertilizers because nutrients are released gradually through decomposition and mineralization^{15,16}. Although several studies have reported positive effects of organic inputs on aboveground responses such as biomass production, nutrient uptake and yield. However, studies on organic and/or natural inputs on spatial and temporal distribution of root traits and nutrients acquisition in plants parts are essential but hardly reported in pigeonpea crop. The invisible nature of root through naked eye, but interconnected all over the heterogeneous soil mass making them extremely challenging to separate out or investigate in-situ, it may be one of the reasons to studied less.

The root system distribution in the soil layers is influenced by various factors like quantity, timing of fertilizers application, irrigation interval, soil properties, tillage management, and cultivars^{17,18}. The accumulation of fertilizers into soil increased root growth, as evidenced by an increase in root morphological traits such as root length, root surface area and other traits. Additionally, several nutrients can influence root shape and architecture¹⁹ and in specific, the level of nutrients availability may promote or prevent the growth of various root morphological traits. These suggest that the root traits of various soil layers show a significant role to define crop grain yield. The application of cow-based solid and liquid manures enhanced root length and grain yield over inorganic fertilizer^{20–22}. As such, compost application in grapevines resulted in significantly root density (52 roots/m²) and yield (15–24%) over control²³. The enhanced root traits, such as root length, volume, surface area and biomass, improved P and K acquisition from soil layers and resulted in higher grain yield^{24,25}. Moreover, the higher root length density (RLD) and root weight density were observed in 0–20 cm soil depth than the 20–40 cm soil depth at the flowering stage²⁶. Additionally, regression analysis showed positive relationships between grain yield and root weight density at 0–20 cm ($R^2 = 0.43$, $P < 0.05$), as well as between grain yield and RLD in 20–40 cm soil depth ($R^2 = 0.34$), indicated the importance of root distribution across soil depths for yield formation. In this light, compost and natural farming offer viable alternatives. Moreover, the mechanistic relationships linking root traits with N, P and K acquisition and subsequent seed yield of pigeonpea crop under semiarid field conditions are still poorly understood.

Therefore, it is hypothesized that crop management practices regulate nutrient acquisition and grain yield in pigeonpea by altering the spatial and temporal distribution of root systems. To test this hypothesis, a field experiment was conducted using graded compost applications and natural farming practices to examine their influence on root development and nutrient uptake dynamics. Therefore, this study aimed to: (i) study depth-wise root morphological traits (0–15 and 15–30 cm soil layers) at key physiological growth stages of pigeonpea and (ii) evaluate the contribution of root traits to N, P, and K acquisition and their relationship with seed yield. By integrated field based assessment of root functional traits with nutrient acquisition dynamics and grain yield of pigeonpea, the study identifies key root traits and nutrients as mechanistic drivers of yield in pigeonpea-wheat based cropping system under semiarid conditions. These findings will provide mechanistic insights into how nutrient management through natural resources shape root system development and nutrient acquisition, offering a scientific basis for designing root-oriented strategies to improve productivity and sustainability of

crops. The output of this study will contribute to these two crucial sustainable development goals, viz. “Zero hunger” and “Life on land” on a larger scale.

Materials and methods

Study site

A field trial was initiated with pigeonpea and wheat crop rotation at the research farm of ICAR- Indian Agriculture Research Institute (IARI), New Delhi, India, during 2021–22. The study area is located at 28°38'N latitude and 77°10'E longitude at an elevation of 248 m above mean sea level. The experimental soil represents the north-western part of the Indo-Gangetic plain (IGP) and is classified as Typic Haplustepts belonging to the Holombi series. The soil was sandy loam texture with 71.7%, 12% and 16.3%, sand, silt, and clay content, respectively. Additionally, analysis of soil showed available nitrogen (188 kg ha⁻¹), available phosphorus (13.8 kg ha⁻¹), available potassium (173 kg ha⁻¹), and organic carbon (0.42%), pH (8.10) at the starting time of field experiment. At pigeonpea harvest in 2024, the soil exhibited improved fertility status, with available nitrogen, phosphorus and potassium increasing to 213, 23.1 and 458 kg ha⁻¹, respectively, while organic carbon content increased to 0.69% and soil pH decreased to 7.63. The experimental site is characterised by semi-arid climate and hot summers. The average maximum and minimum temperatures were 37.4 and 28 °C in July (sowing time), and 29.5 and 14.9 °C in November (Harvesting time) (Fig. 1). During the years 2024–25, the total rainfall of the area was 902.6 mm. However, rainfall was unevenly distributed across the crop growth stages (Fig. 2).

Experimental design and crop management

The field trial was started in 2021–22 with pigeonpea-wheat cropping system in a randomized block design (RBD) with seven treatments and three replications. The treatments consisted T₁-control (no fertilizer and manure), T₂-conventional farming (CF), T₃-natural farming practices, T₄-compost (6 t ha⁻¹), T₅-compost (12 t ha⁻¹), T₆-compost (18 t ha⁻¹), T₇-compost (24 t ha⁻¹). The recommended dose of fertilizer N: P₂O₅: K₂O was at the rate of 20:60:20 kg ha⁻¹. The individual plot size was 5.6 m wide and 5.0 m long. The study was conducted during the 2024–2025 growing season with pigeonpea crop. After harvesting the wheat crop, compost was spread on the surface in accordance with the experimental treatments across all corresponding experimental units about 7 days before sowing the pigeonpea crop. Then, the land was prepared, and the soil was ploughed to ensure thorough mixing of the organic amendments after incorporation. Seeds were sown in the first week of July in the cropping season 2024–25. The seeds were sown manually in rows, with a 10 × 15 cm spacing, at a depth of about 5 cm. The half dose of nitrogen and full dose of phosphorus and potassium, through urea, diammonium phosphate, and muriate of potash, respectively, were provided at sowing time, and the remaining nitrogen fertilizer was supplied by two split applications at different crop stages (30DAS & 60DAS).

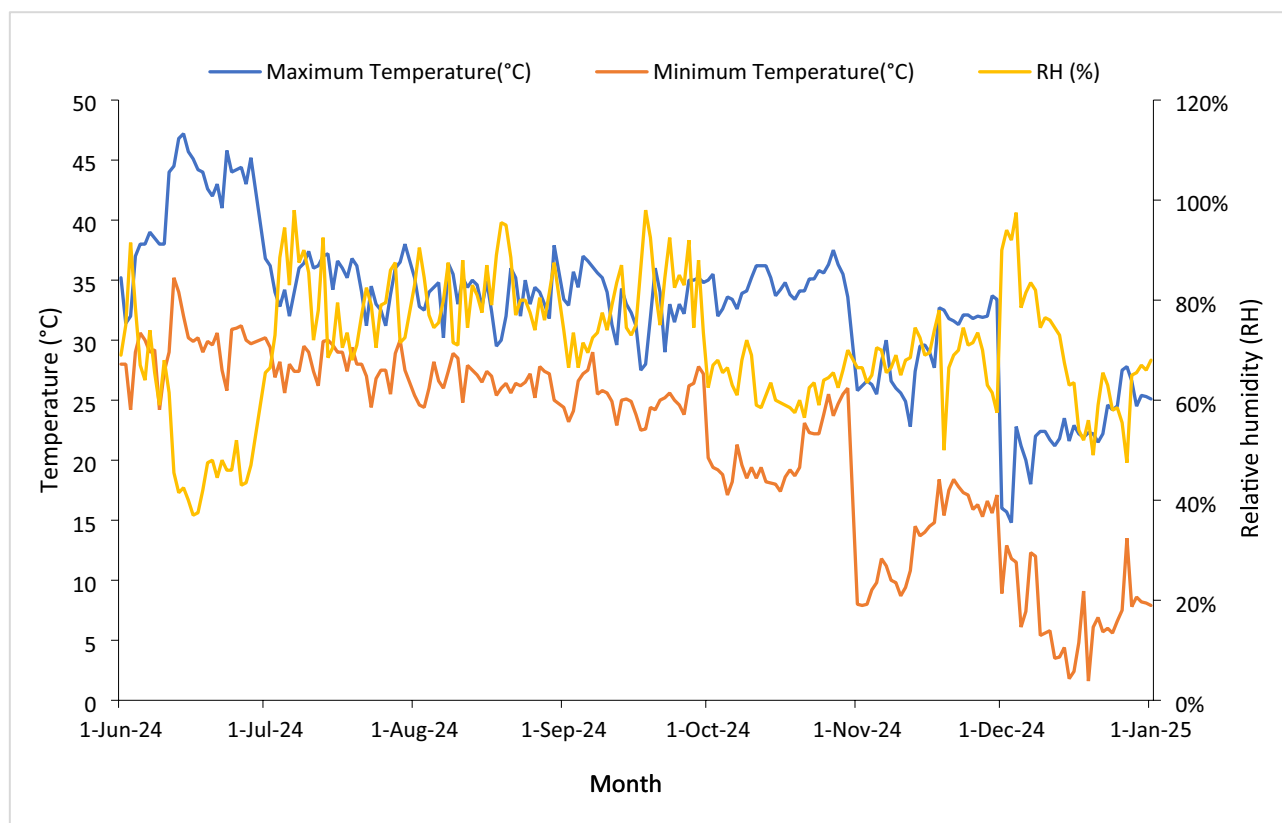


Fig. 1. Weather conditions of the site of experiment during the year 2024–2025.

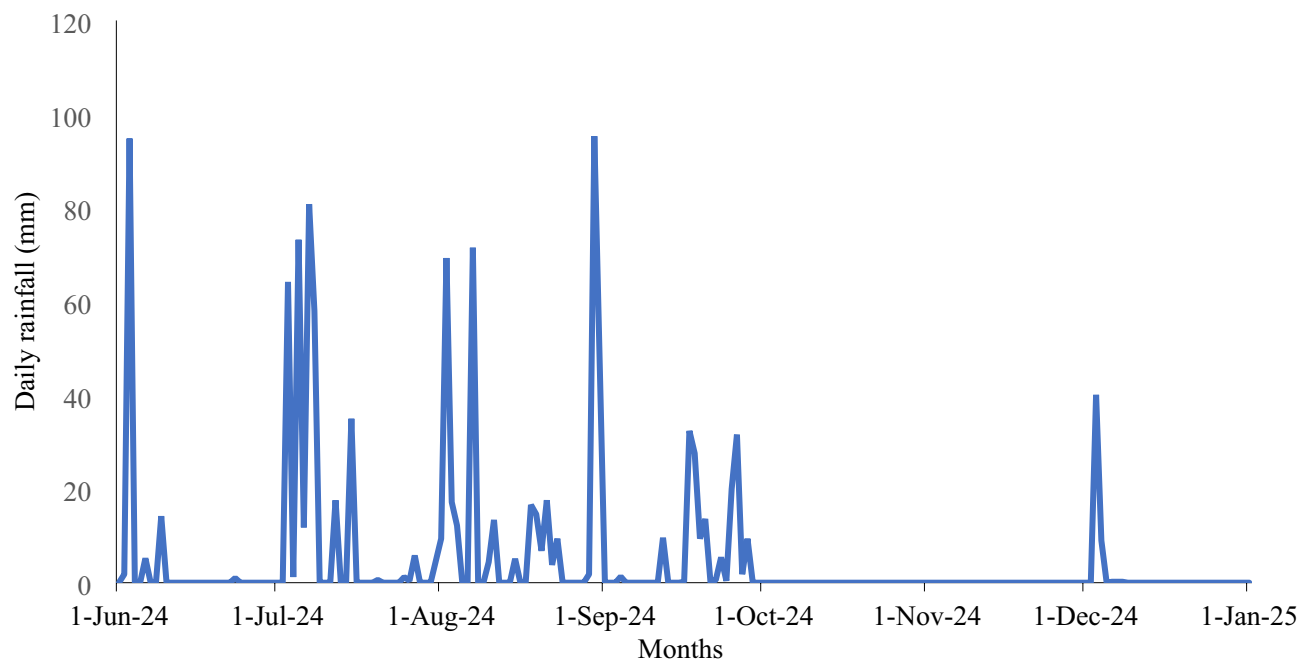


Fig. 2. Daily rainfall (mm) distribution of the site of experiment during the year 2024–2025.

Among various inputs of natural farming practice, mulching in pigeonpea crop with wheat crop straw (5t ha⁻¹) immediately after sowing and seed treatment with *beejamrit*. Ghanjeevamrit was supplied (250 kg ha⁻¹) as basal dose and foliar application of *panchagavya* at the rate 3% and *Jeevamrit* at the rate 500 L ha⁻¹ at 30, 60, and 90 days after sowing in the pigeonpea crop. The details of nutrient contents in natural farming amendments and compost are represented in Table S1. Except for the 3–4 manual weeding, no other management was included in the control treatment. Weed management practices varied according to agricultural management systems. NF plots avoid hand weeding in favour of mulching with wheat straw when plant height was 25–30 cm. The plots with compost treatments and CF application employed hand weeding at 30, 45, and 60 DAS. The crop was harvested manually 5 cm above the ground at physiological maturity of crops in last week of November.

Root sample collection

To study the morphology of root behaviour in response to different fertilizer management the root sample was collected at 45 (vegetative growth stage), 90 (flowering stage), and 135 (pod filling stage) days after sowing (DAS). For collection of root samples, five plant roots per plot (total of 15 per treatment) were randomly selected at each sampling stage and excavated from two soil depths (0–15 cm and 15–30 cm). Root samples were collected using a “semi-automatic rhizospheric soil and crop root sampling device” (16 cm internal diameter and 40 cm height). It was placed concentric to the base of the plant stem and pushed into the soil with a hand hammer. The core with undisturbed pigeonpea root system was hollowed out. These samples were brought to the laboratory, and the roots were rinsed three to four times with tap water and twice with distilled water using 2 mm and 0.5 mm sieves to eliminate soil particles. An additional 0.2 mm sieve was also used to ensure that no fine root material was gone. Organic debris and dead roots were manually removed from vital roots based on their color, shape, and flexibility. All visible roots were collected, and then paper towels were used to remove excess water; subsequently, fresh weight was taken. The roots collected at each depth were placed in a self-sealing polythene bag (12.5- μ m thickness) and the root samples were stored in the refrigerator at 4 °C immediately^{27,28}

Root image acquisition and processing using WinRHIZO™

To quantify root morphological traits, roots were placed in acrylic trays (30 cm × 20 cm) with a 3–4 mm deep layer of water on the scanner. This allows the roots to be arranged in a suitable manner to decrease overlap and root crossing, and to be aligned vertically on the scanning plates. Thereafter, roots were scanned with a high-resolution root scanner (EPSON Expression 11000 XL, Japan), resulting in an 8-bit grayscale image. WinRHIZO™ software was used for various root measurements, including root length (RL), root surface area (RSA), and root volume (RV).

Root length density (RLD)

Root length density (cm cm^{-3}) was calculated as the ratio of the root length (RL, cm) to the volume of the core (V, cm^3)²⁹.

$$RLD = \frac{\text{Root length (RL)}}{\text{Volume of core}} \quad (1)$$

Relative root growth constant

The relative root growth constant (k) represents increase in root length over given time interval relative to the initial root length. It was calculated using the following expression³⁰.

$$\text{Relative root growth constant } (k, \text{ cm day}^{-1}) = \left[\frac{\ln RL_2 - \ln RL_1}{t_2 - t_1} \right] \quad (2)$$

Here, RL_1 and RL_2 represents root length at times t_1 and t_2 , with subscripts 1 and 2 referring to the first and second observations, respectively.

Specific root length

Specific root length (SRL) is most important morphological parameter of fine roots. The data recorded at 45, 90 and 135 DAS were also used to compute the SRL. The SRL is the ratio of length to mass of the root fragments³¹.

$$\text{Specific root length } (\text{m g}^{-1}) = \frac{\text{Root length (RL)}}{\text{Root dry weight (RDW)}} \quad (3)$$

The maximum plants observed with larger growth of absorbing root were in the upper part of the root zone (0–15 cm) and near the base of the plant.

Nutrient content analysis

The total nitrogen (N), phosphorus (P), potassium (K) concentrations were estimated at 30 cm root depth under direct observation. At each stage, the sampled roots were oven-dried at 70⁰ until constant weight and the root dry weight was measured for each depth. After being weighed, root samples were crushed to pass through a 0.5 mm sieve, then digested with nitric acid (HNO_3) and perchloric acid (HClO_4) (10:4) to determine total phosphorus (P) and potassium (K) concentrations. Total P content was measured with the vanadomolybdate method³² and total K content with the flame photometer method³³ from the P and K concentrations in the digest, respectively. Nitrogen (N) was determined using CHNS analyser³⁴. The N, P and K uptake were calculated by multiplying each nutrient content by the root dry biomass³⁵.

Statistical analysis

The data were statistically analysed using the analysis of variance (ANOVA) technique as applicable to randomized complete block designs³⁶. First, all data sets were analysed considering all growth stages as a source of variation in addition to the treatments. The main effect of the growth stage and the interaction effect of stage $\times$ treatments on grain yield and root morphological traits were determined using Duncan's multiple range test (DMRT) among the treatment means at the 5% probability level by using Python, version 3.11. correlation among root traits, nutrient uptake and yield was performed using Python, version 3.11. Relationship between root traits and seed yield (mean value of selected root traits measured at 0–30 cm soil depth across the physiological growth stages were averaged) subjected to multiple regression analysis (SPSS version 23) to predict the most important root traits governing the yield of pigeonpea.

Results

Spatiotemporal distribution of root morphological traits

Root length

The root length (RL) of pigeonpea in the 0–15 and 15–30 cm layers increased rapidly after germination to vegetative growth (45 DAS) and reached maximum at the flowering stage (90 DAS), and afterwards declined gradually until the pod filling stage (135 DAS) (Fig. 3A, B & C). At 45 DAS in the surface layer (0–15 cm), it varied from 296 cm (control) to 446 cm (compost 12 t ha⁻¹). The NF grown plants (354 cm) recorded significantly ($p < 0.05$) lower root length than compost 6 & 12 t ha⁻¹ (7.8 & 20.6%, respectively), CF (10.8%). In the subsurface layer, compost 12 t ha⁻¹ (185 cm) grown plants recorded highest RL, although it was statistically similar to CF plots but higher than NF and other plots (154 cm). At 90 DAS, root length closely followed the pattern of 45 DAS among treatments in both soil depths, although the intensity of variation was different. The NF plots (453 cm) in surface layer recorded significantly ($p < 0.05$) 7.4%, 9.0% and 19.0% lower RL than NPK, compost (6 t ha⁻¹) and compost (12 t ha⁻¹) plots, respectively. In the subsurface layer (15–30 cm), root length ranged from 196 cm to 239 cm and it was highest in compost 12 t ha⁻¹, followed by compost 6 t ha⁻¹ (230 cm), CF (227 cm) and NF (211 cm) plots.

Root surface area

At 45 DAS, the root surface area (RSA) of pigeonpea in the topsoil layer (0–15 cm) was significantly higher ($p < 0.05$) in the NF plots compared to the control, and recorded an increase of 33.1% (Table 1). The RSA in NF plots was statistically comparable to those with CF and compost at 6 t ha⁻¹, and it was lower than the plots with 12 t ha⁻¹ compost, which showed an increase of 18.7%. In the subsurface layer, RSF in NF plots revealed a comparable pattern as like surface layer. Similar to root length, RSA was recorded highest at the flowering stage (90 DAS) across both depths. At 90 DAS, in surface layer, the order of RSA was: compost (12 t ha⁻¹) > compost (6 t ha⁻¹) = NPK = NF > compost (18 t ha⁻¹) = compost (24 t ha⁻¹) > control. The RSA in compost (12 t ha⁻¹) plots was 17.6% and 23.0% higher than CF and NF plots, respectively. In the subsurface layer, NF plots had significantly ($p < 0.05$) lower values than compost (6 & 12 t ha⁻¹) (13.0–13.3%) and CF (13.0%) plots. However, it

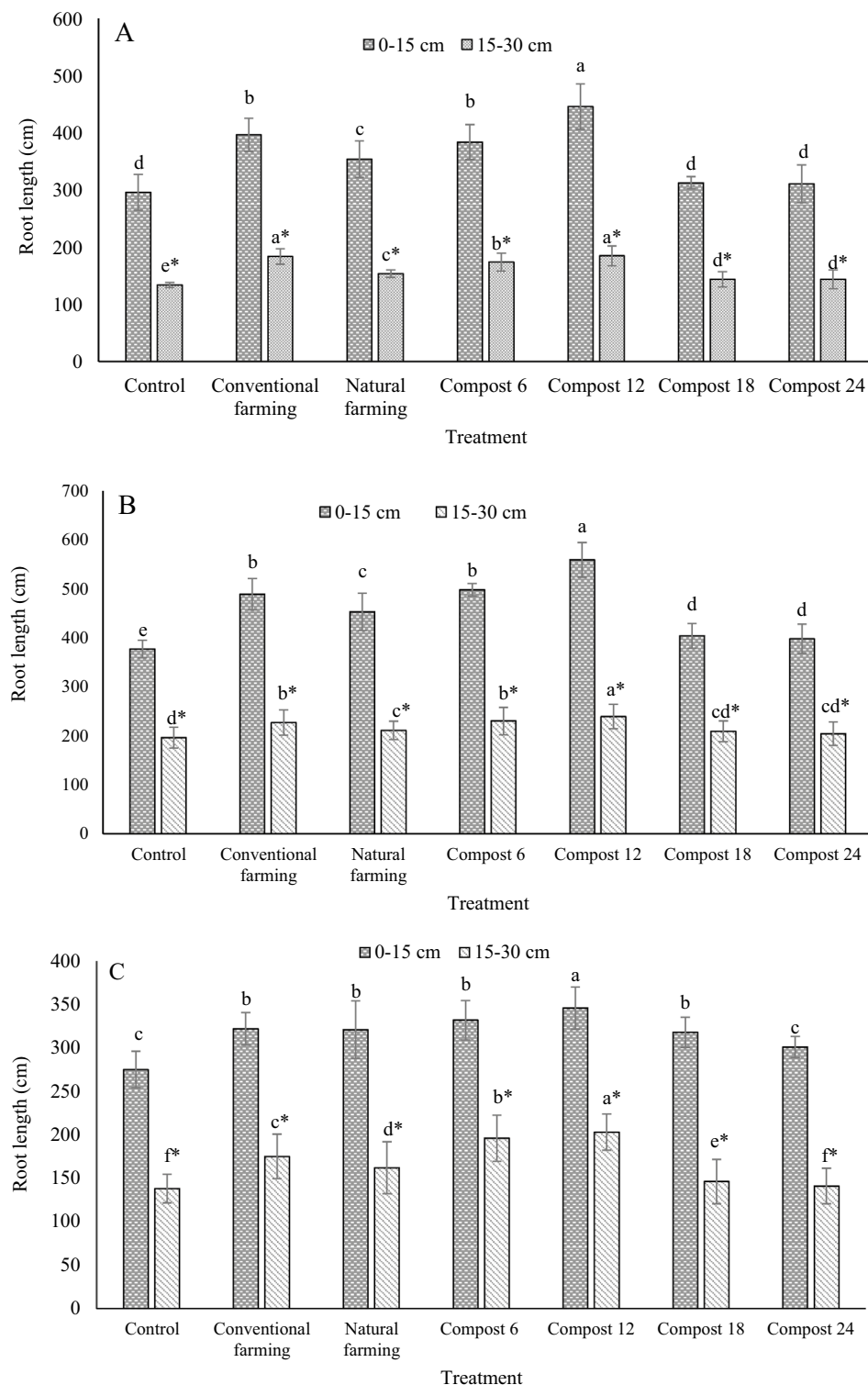


Fig. 3. Root length of pigeonpea at 45 days after sowing (DAS) (A), 90 DAS (B) and 135 DAS (C) in 0–15 and 15–30 cm soil depths. Error bars indicate standard error of mean ($\pm$). Values with the same letter above the columns indicate no significant differences at $p=0.05$ as per Duncan's multiple range test (DMRT). Compost 6- compost 6 t ha⁻¹; compost 12- compost 12 t ha⁻¹; compost 18 - compost 18 t ha⁻¹; compost 24 – compost 24 t ha⁻¹.

Treatments	Surface area (cm ²)						Root volume (cm ³)					
	45 DAS		90 DAS		135DAS		45 DAS		90 DAS		135DAS	
	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm
Control	58.0 ^d	27.6 ^c	105 ^d	49.8 ^c	87.4 ^d	37.4 ^e	0.95 ^d	0.68 ^d	2.33 ^e	1.22 ^c	2.21 ^d	0.92 ^c
Conventional farming	87.9 ^{ab}	38.2 ^a	159 ^b	66.2 ^a	121 ^{ab}	52.9 ^{bc}	1.48 ^{ab}	0.94 ^a	4.12 ^c	1.64 ^a	3.61 ^b	1.34 ^{abc}
Natural farming	77.2 ^c	34.7 ^{ab}	152 ^b	57.6 ^b	121 ^{ab}	46.7 ^{cd}	1.35 ^b	0.86 ^{ab}	4.06 ^c	1.39 ^b	3.63 ^{ab}	1.15 ^{bc}
Compost (6 t ha ⁻¹)	82.4 ^{abc}	38.0 ^a	163 ^b	66.2 ^a	123 ^{ab}	55.8 ^{ab}	1.39 ^b	0.93 ^a	4.25 ^b	1.66 ^a	3.63 ^{ab}	1.39 ^{ab}
Compost (12 t ha ⁻¹)	94.9 ^a	38.4 ^a	187 ^a	66.4 ^a	128 ^a	63.3 ^a	1.61 ^a	0.94 ^a	4.98 ^a	1.67 ^a	3.77 ^a	1.58 ^a
Compost (18 t ha ⁻¹)	66.3 ^{bd}	32.0 ^b	118 ^c	56.2 ^b	113 ^{bc}	42.2 ^{de}	1.11 ^c	0.80 ^{bc}	2.89 ^d	1.39 ^b	3.20 ^c	1.05 ^{bc}
Compost (24 t ha ⁻¹)	62.8 ^d	30.9 ^{bc}	116 ^{cd}	52.6 ^{bc}	105 ^c	39.5 ^{de}	1.04 ^c	0.77 ^{cd}	2.69 ^d	1.31 ^{bc}	2.92 ^c	0.99 ^{bc}

Table 1. Effect of natural farming and compost on root surface area (cm²) and root volume (cm³) at different crop growth stages of pigeonpea. Results sharing the same letter indicate no significant differences at $p=0.05$ (Duncan's multiple-range test) within same column.

was similar to compost (18 & 24 t ha⁻¹) and higher than control (15.7%) plots. The RSA continuously declined to 135 DAS across all treatments in all soil depths.

Root volume

The root volume (RV) exhibited a consistent spatiotemporal variation across treatments in different stages, declining with increasing soil depth (Table 1). At 45 DAS, RV in the 0–15 cm layer ranged from 0.95 to 1.61 cm³, and all treatments recorded significantly higher values than the control (0.95 cm³). The highest RV was observed in compost 12 t ha⁻¹ (1.61 cm³), which was statistically comparable to CF ($p<0.05$) and significantly higher than the other treatments. In the 15–30 cm layer at 45 DAS, compost (6 & 12 t ha⁻¹), NPK, and NF plots exhibited statistically similar RV values ($p<0.05$). At 90 DAS (flowering stage), RV reached its highest value in the topsoil layer (0–15 cm), following the sequence among treatments: compost (12 t ha⁻¹) > compost (6 t ha⁻¹) > CF = NF > compost (18 t ha⁻¹) = compost (24 t ha⁻¹) > control. The percentage increase in RV for compost at 12 t ha⁻¹ plots compared to compost at 6 t ha⁻¹, NPK, and NF plots was 17.1%, 20.9%, and 22.7%, respectively. At 90 DAS in the subsurface layer, the RV in NF plots (1.39 cm³) was less than that in compost at 6 & 12 t ha⁻¹ (1.66 & 1.67 cm³) and CF plots (1.64 cm³), but it was statistically comparable to compost plots (8 & 24 t ha⁻¹).

Fresh weight of root

The fresh weight of root (FW) at 45 DAS in top soil layer (0–15 cm) was highest under compost at 12 t ha⁻¹ (3.17 g) and lowest in control plots (2.02 g) (Fig. 4A). In 0–15 cm, application of compost (12 t ha⁻¹) led to increase in FW by 16.5% and 12.4% when compared to NF and compost at (6 t ha⁻¹), respectively. In the subsurface layer (15–30 cm), the NF plots (0.15 g) had statistically comparable FW to those of the compost (6, 18 & 24 t ha⁻¹) plots, but were lower than the compost 12 t ha⁻¹ (28.6%) and CF (21.1%) plots. At 90 DAS, root FW in the surface layer increased over 45 DAS, and ranged from 6.94 g (control) to 13.2 g (compost 12 t ha⁻¹) (Fig. 4B). NF plots, with a weight of 10.2 g, were statistically comparable to CF plots, but lower than compost treatments of 6 and 12 t ha⁻¹, which was 15.0% and 22.7% higher, respectively. At 90 DAS in the sub-surface layer, the compost treatment of 12 t ha⁻¹ recorded the highest value of 1.11 g, followed by the 6 t ha⁻¹ compost treatment at 1.06 g, with no significant difference observed between these two treatments. Afterwards, root fresh weight decreased from 90 to 135 DAS (Fig. 4C).

Root dry weight

Root dry weight (RDW) demonstrated differential performance across depth and stages (Fig. 5A, B & C). At 45 DAS, the RDW in plots where 12 t ha⁻¹ of compost was applied was higher at both soil depths (0–15 cm and 15–30 cm) compared to other treatments, with the exception of plots treated with 18 and 24 t ha⁻¹ of compost. At 90 DAS in the surface layer, the NF plots recorded a root dry weight of 4.15 g, that was similar to compost (6 t ha⁻¹) and CF plots, and higher than those with larger compost applications of 18 and 24 t ha⁻¹ (3.82 and 3.71 g, respectively) as well as the control plots (3.68 g). However, it was lower than the plots with 12 t ha⁻¹ compost (4.68 g). In the subsurface layer at the same stage, the sequence of RSW was: compost (12 t ha⁻¹) = CF > compost (6 t ha⁻¹) = NF > compost (18 t ha⁻¹) = compost (24 t ha⁻¹) > control.

Root length density

At 45 days after sowing (DAS), root length density (RLD) in the 0–15 cm soil layer was significantly higher under compost application at 12 t ha⁻¹, which was 12.1% and 26.5% higher compared with the CF and NF treatments, respectively (Table 2). In contrast, the NF plots exhibited significantly ($p<0.05$) higher RLD, by 12.5%, 13.6%, and 19.3%, relative to compost applied at 18 and 24 t ha⁻¹ and the control plots, respectively. In the subsurface layer (15–30 cm) at 45 DAS, RLD was comparatively higher under CF and compost at 12 t ha⁻¹ than other treatments. At 90 DAS (flowering stage), RLD attained its maximum values again under compost application at 12 t ha⁻¹, measuring 1.85 mm cm⁻³ in the surface layer and 0.79 mm cm⁻³ in the subsurface layer. In the 0–15 cm layer, compost at 12 t ha⁻¹ and CF treatments increased RLD by 23.3% and 8.0%, respectively, compared with the NF treatment. In the 15–30 cm layer at 90 DAS, RLD followed the order: compost (12 t ha⁻¹)

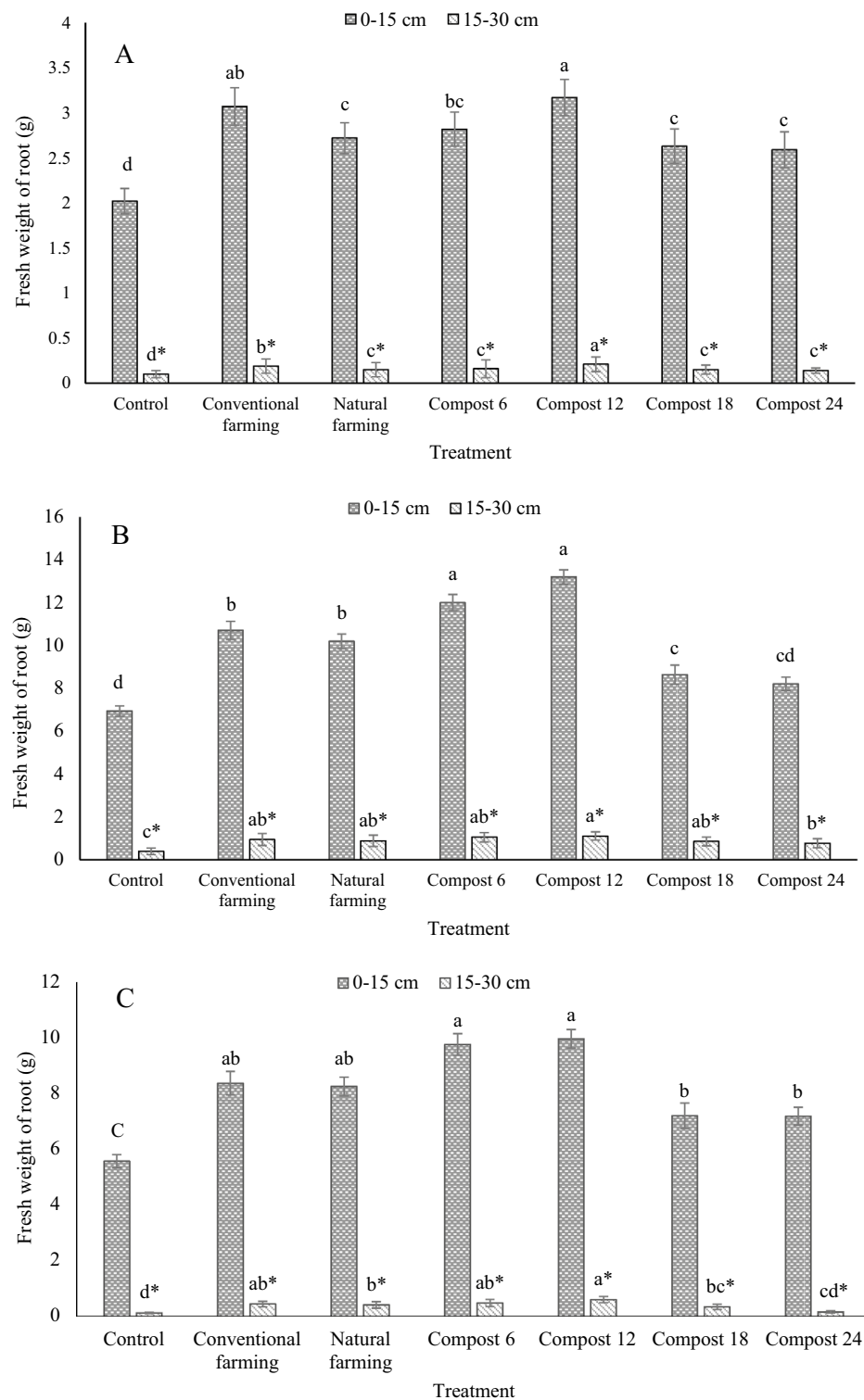


Fig. 4. Fresh weight of root (g) at 45 days after sowing (DAS) (A), 90 DAS (B) and 135 DAS (C) in 0–15 and 15–30 cm soil depths. Error bars indicate standard error of mean ($\pm$). Values with the same letter above the columns indicate no significant differences at $p=0.05$ as per Duncan's multiple range test (DMRT). Compost 6 - compost 6 t ha⁻¹; compost 12 - compost 12 t ha⁻¹; compost 18 - compost 18 t ha⁻¹; compost 24 - compost 24 t ha⁻¹.

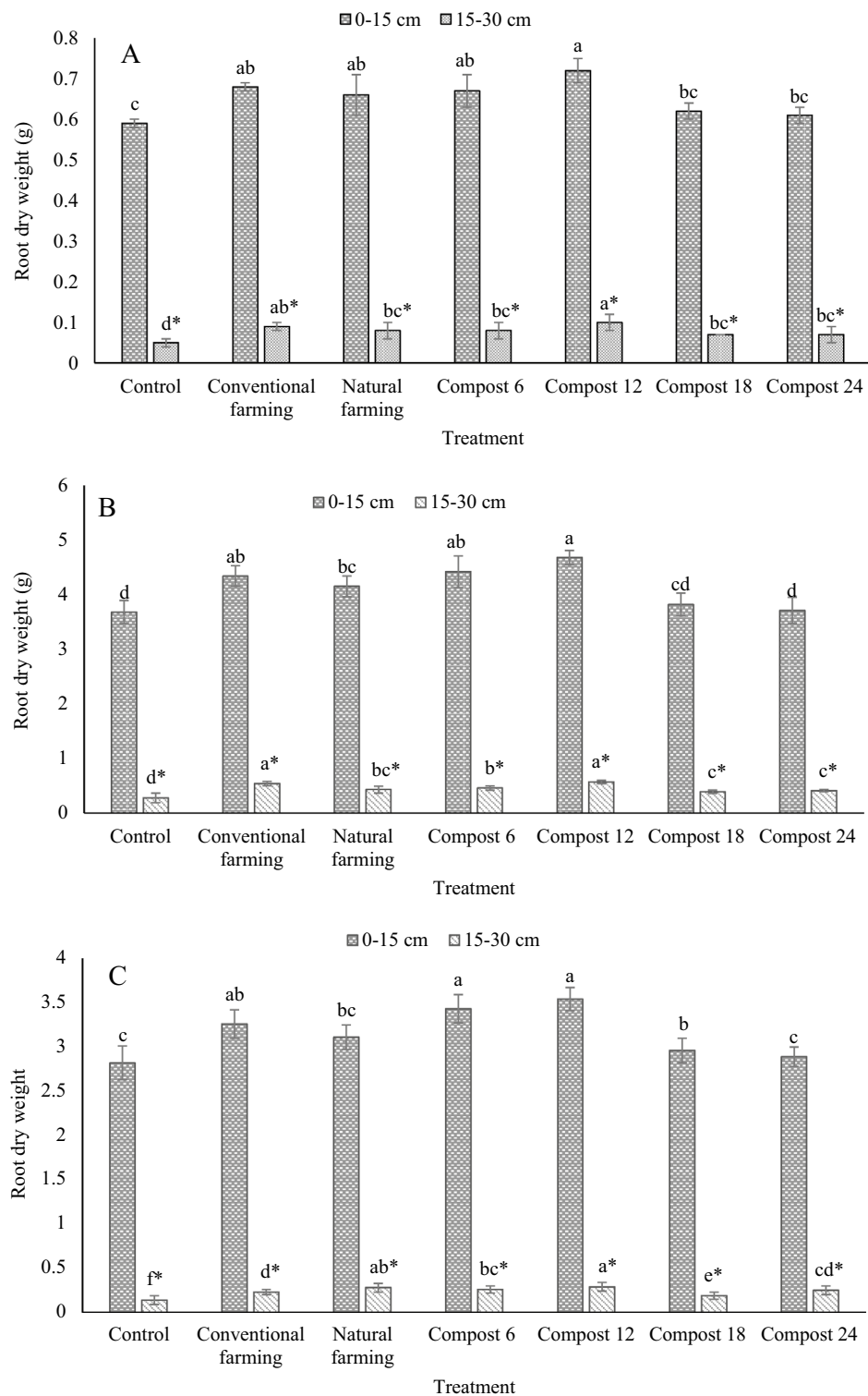


Fig. 5. Root dry weight of pigeonpea at 45 days after sowing (DAS) (A), 90 DAS (B) and 135 DAS (C) in 0–15 and 15–30 cm soil depths. Error bars indicate standard error of mean ($\pm$). Values with the same letter above the columns indicate no significant differences at $p=0.05$ as per Duncan's multiple range test (DMRT). Compost 6 - compost 6 t ha⁻¹; compost 12 - compost 12 t ha⁻¹; compost 18 - compost 18 t ha⁻¹; compost 24 - compost 24 t ha⁻¹.

Treatments	Specific root length (m g ⁻¹)						Relative root length density (mm cm ⁻³)					
	45 DAS		90 DAS		135DAS		45 DAS		90 DAS		135DAS	
	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm
Control	5.05 ^d	26.4	1.06	7.29 ^a	0.98	10.9 ^a	0.98 ^d	0.44 ^e	1.25 ^e	0.65 ^d	0.91 ^c	0.46 ^f
Conventional farming	5.91 ^{ab}	19.7	1.11	4.83 ^b	0.98	7.37 ^b	1.32 ^b	0.61 ^a	1.62 ^b	0.75 ^b	1.07 ^b	0.58 ^c
Natural farming	5.32 ^{bcd}	20.4	1.09	5.00 ^b	1.03	5.77 ^b	1.17 ^c	0.51 ^c	1.50 ^c	0.70 ^c	1.06 ^b	0.54 ^d
Compost (6 t ha ⁻¹)	5.74 ^{abc}	22.2	1.16	4.24 ^b	0.99	7.53 ^b	1.27 ^b	0.58 ^b	1.65 ^b	0.76 ^b	1.10 ^b	0.65 ^b
Compost (12 t ha ⁻¹)	6.22 ^a	19.4	1.20	4.23 ^b	0.98	7.07 ^b	1.48 ^a	0.61 ^a	1.85 ^a	0.79 ^a	1.15 ^a	0.67 ^a
Compost (18 t ha ⁻¹)	5.14 ^{cd}	22.0	1.07	5.34 ^b	1.09	7.67 ^b	1.04 ^d	0.48 ^d	1.34 ^d	0.69 ^{cd}	1.05 ^b	0.48 ^e
Compost (24 t ha ⁻¹)	5.04 ^d	21.9	1.08	4.97 ^b	1.05	5.67 ^b	1.03 ^d	0.48 ^d	1.32 ^d	0.68 ^{cd}	1.00 ^c	0.47 ^f

Table 2. Effect of natural farming and compost on specific root length (m g⁻¹) and root length density (mm cm⁻³) at different crop growth stages of pigeonpea. Results sharing the same letter indicate no significant differences at $p = 0.05$ (Duncan’s multiple-range test) within same column.

> compost (6 t ha⁻¹) = CF > NF > compost (18 t ha⁻¹) = compost (24 t ha⁻¹) > control, with compost at 12 t ha⁻¹ recording RLD values 7.01% and 12.8% higher than those under CF and NF treatments, respectively.

Specific root length and root growth constant

The highest specific root length (SRL) of pigeonpea in the 0–15 cm soil layer was recorded at 45 DAS, after which a steady decrease was observed (Table 2). In the 15–30 cm soil layer, SRL also peaked at 45 DAS, decreased up to 90 DAS, and subsequently increased again by 135 DAS. At 45 DAS in the surface layer, the highest SRL was recorded under compost 12 t ha⁻¹ (7.66 m g⁻¹), which was notably 13.4% higher than the NF plots. In contrast, the control plots recorded relatively higher values in the 15–30 cm soil layer across all stages. However, the differences between treatments were not significant ($p < 0.05$) at 45 DAS, while significant effects were noted at 90 and 135 DAS. The root growth constant (RGC) continues to decline in both depth (0–15 cm and 15–30 cm) from 45 to 135 DAS (Table S2). The highest value of RGC was observed at 45 DAS, in the surface layer under compost 12 t ha⁻¹ (0.136 cm day⁻¹), followed by CF (0.133 cm day⁻¹), compost 6 t ha⁻¹ (0.132 cm day⁻¹) and NF plots (0.130 cm day⁻¹). In sub-surface layer at same stage, RGC followed the order: compost (12 t ha⁻¹) > CF = compost (6 t ha⁻¹) > NF > compost (18 t ha⁻¹) = compost (24 t ha⁻¹) = control plots.

Temporal variations of pigeonpea root nutrients

Nutrient

Nitrogen (N) content in root varied from 0.54 to 3.96% across different physiological growth stages of pigeonpea. Natural farming consistently recorded higher total N content throughout the crop growth stages (Fig. 6A). At 45 DAS, N content of root under NF was statistically similar to NPK, compost (6 & 12 t ha⁻¹), and significantly higher than compost (18 & 24 t ha⁻¹) and control plots grown plants. Among the different crop growth stages, maximum N content under NF (3.96%) was observed at flowering stage (90 DAS), and it was higher than CF plots by 38.5%. The root phosphorus (P) at 45 DAS under NF plots was significantly higher than the control (55.5%) plots, while similar to NPK, compost (6, 18 & 24 t ha⁻¹) plots, (Fig. 6B). The P content in root was maximum at flowering stage followed by vegetative growth (45 DAS) and pod filling stage (135 DAS). At the flowering stage (90 DAS), the phosphorus content in the root was highest with compost application at 12 t ha⁻¹ (0.35%), followed by compost at 6 t ha⁻¹ (0.30%), CF (0.29%), and NF (0.22%) plots. Similar to P content, at 45 DAS, the compost treatment at 12 t ha⁻¹ exhibited highest potassium content, which was significantly higher than the other treatments, showed an increase of 12.6%, 45.9%, and 62.0% compared to the NPK, NF, and compost at 18 t ha⁻¹ plots, respectively (Fig. 6C). At 90 DAS, the NF plots showed significantly comparable K content in root to those of CF and compost at 18 t ha⁻¹ plots. However, it was 28.0% and 48.9% higher than the compost at 24 t ha⁻¹ and the control plots, respectively, yet remained 24.5% and 14.1% lower than the compost at 12 t ha⁻¹ and 6 t ha⁻¹, respectively. Subsequently, the nutrient levels (N, P, K) decreased until 135 DAS across all treatments. The improved nutrient content with NF and compost applications at 6 and 12 t ha⁻¹ suggested a more effective alignment of nutrient release with the crop’s needs compared to higher compost applications or the use of inorganic fertilizers alone.

Nutrients uptake by root

The nitrogen (N) uptake ranged from 4.59 to 201 mg plant⁻¹ across all growth stages (Table 3). At 45 DAS, the compost application at 12 t ha⁻¹ resulted in the highest N uptake, surpassing the uptake levels observed with compost applications of 18 and 24 t ha⁻¹ by 24.4% and 52.8%, respectively, as well as the control plots by 132%. At 90 DAS, N uptake followed the order: compost (12 t ha⁻¹) = compost (6 t ha⁻¹) > NF > compost (18 t ha⁻¹) = CF > compost (24 t ha⁻¹) > control plots. The NF plots recorded 29.2% higher uptake than CF plots, but 11.0% lower uptake than compost (12 t ha⁻¹) plots. Phosphorus uptake, at 45 DAS, was significantly highest under compost 12 t ha⁻¹ (1.45 mg plant⁻¹), followed by CF (1.21 mg plant⁻¹) and NF plots (1.09 mg plant⁻¹). At 90 DAS, the NF plots showed higher P uptake compared to the compost plots at 18 and 24 t ha⁻¹ and control plots, with an increase of 36.2%, 51.3%, and 82.9%, respectively. However, these values were still 25.4% and 15.4% lower than those of the compost (12 t ha⁻¹) and CF plots, respectively. The potassium content (K) was significantly ($p < 0.05$) higher than control in all treatments except compost (24 t ha⁻¹) at all growth stages. Throughout all

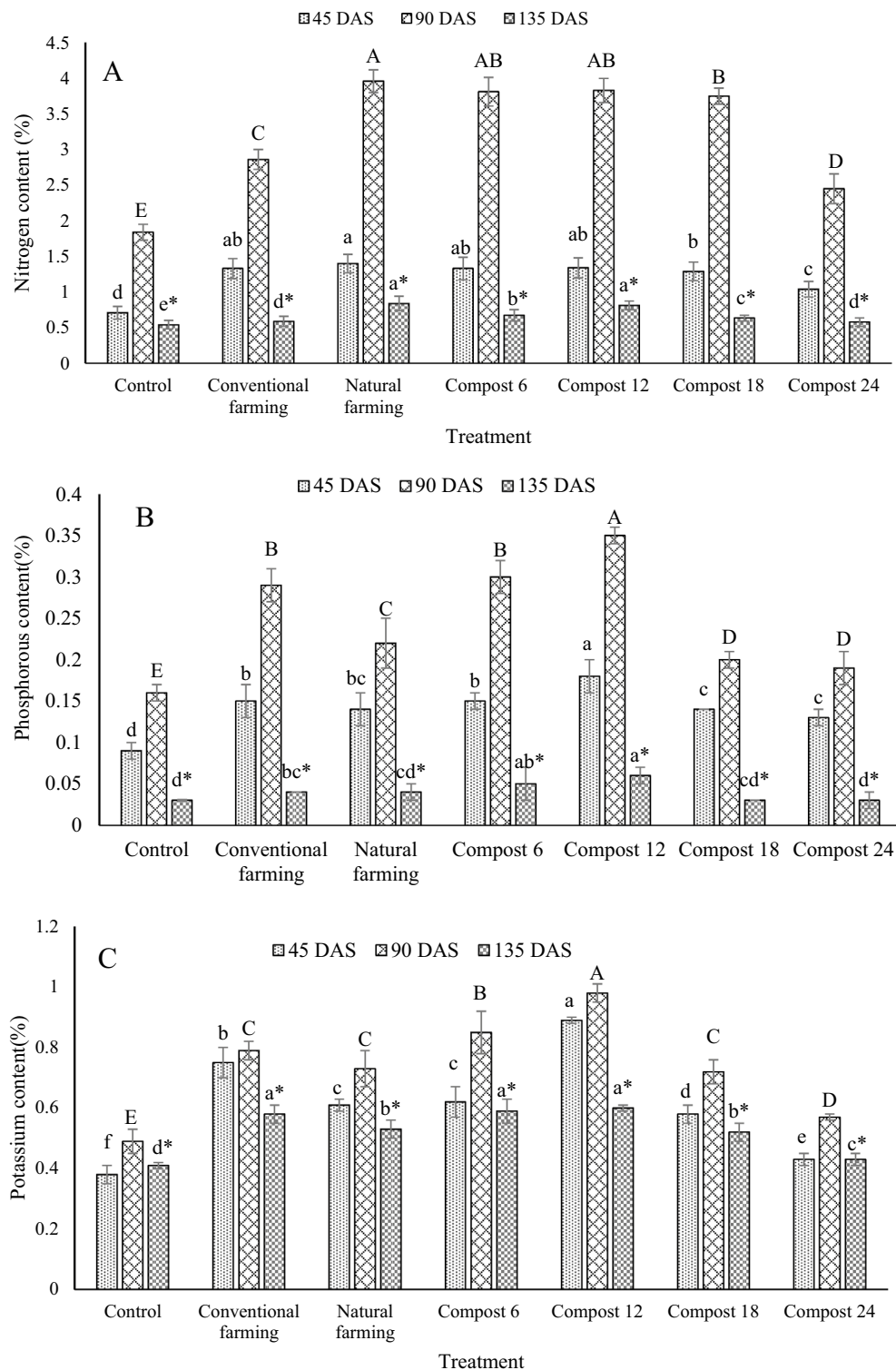


Fig. 6. Nitrogen (A), phosphorous (B) and potassium (C) content in roots at 45, 90 and 135 days after sowing in 0–30 cm soil depth. Error bars indicate mean $\pm$ standard error. Values with the same letter indicate no significant differences at $p=0.05$ as per Duncan's multiple range test (DMRT). Compost 6- compost 6 t ha⁻¹; compost 12- compost 12 t ha⁻¹; compost 18- compost 18 t ha⁻¹; compost 24 – compost 24 t ha⁻¹.

growth stages, the compost applied at 12 t ha⁻¹ resulted in the highest potassium uptake, with K content of 7.99, 46.8, and 23.0 mg plant⁻¹ at 45, 90, and 135 DAS, respectively. At 90 DAS, the NF plots were statistically similar to the CF plots ($p < 0.05$), yet it was 27.5% and 39.9% lower values compared to the compost plots at 6 and 12 t ha⁻¹, respectively.

Nutrient uptake (mg plant ⁻¹)	N uptake			P uptake			K uptake		
	45DAS	90DAS	135DAS	45DAS	90DAS	135DAS	45DAS	90DAS	135DAS
Control	4.59 ^d	74.5 ^c	16.0 ^d	0.58 ^d	6.56 ^e	1.03 ^d	3.21 ^d	15.1 ^e	12.2 ^e
Conventional farming	10.3 ^{ab}	140 ^c	21.8 ^{bc}	1.21 ^b	14.2 ^b	1.76 ^{bc}	5.63 ^c	30.1 ^c	20.8 ^{ab}
Natural farming	10.4 ^{ab}	181 ^b	28.5 ^a	1.09 ^{bc}	12.0 ^c	1.38 ^{bcd}	5.43 ^c	28.1 ^{cd}	18.3 ^{bc}
Compost (6 t ha ⁻¹)	10.0 ^{ab}	186 ^{ab}	23.6 ^b	1.17 ^{bc}	14.6 ^{ab}	1.98 ^{ab}	6.60 ^b	38.8 ^b	21.6 ^a
Compost (12 t ha ⁻¹)	10.9 ^a	201 ^a	31.2 ^a	1.45 ^a	16.1 ^a	2.49 ^a	7.99 ^a	46.8 ^a	23.0 ^a
Compost (18 t ha ⁻¹)	8.76 ^{bc}	158 ^c	20.2 ^{bc}	0.96 ^c	8.81 ^d	1.23 ^{cd}	4.92 ^c	24.8 ^d	16.4 ^{cd}
Compost (24 t ha ⁻¹)	7.13 ^c	101 ^d	18.3 ^{cd}	0.95 ^c	7.93 ^{de}	1.15 ^{cd}	3.93 ^d	17.9 ^e	13.5 ^{de}

Table 3. Effect of natural farming and compost on N, P and K uptake different crop growth stages of pigeonpea. Results sharing the same letter indicate no significant differences at $p=0.05$ (Duncan's multiple-range test) within same column.

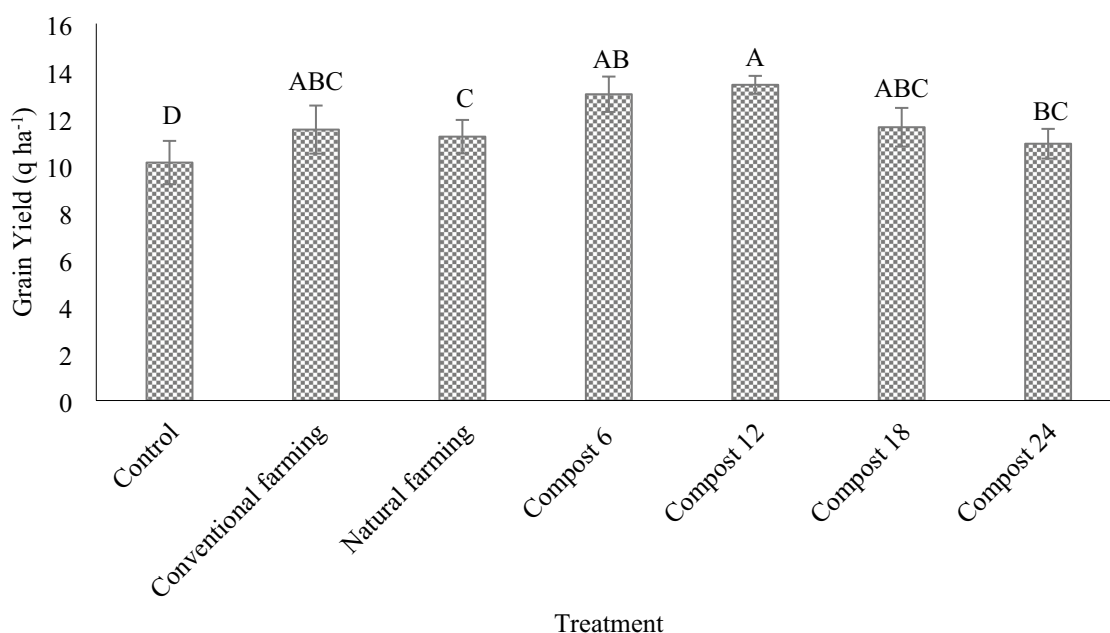


Fig. 7. Effect of crop management strategies on pigeonpea grain yield. Error bars indicate mean $\pm$ standard error. Values with the same letter indicate no significant differences at $p=0.05$ as per Duncan's multiple range test (DMRT). Compost 6 - compost 6 t ha⁻¹; compost 12- compost 12 t ha⁻¹; compost 18 - compost 18 t ha⁻¹; compost 24 - compost 24 t ha⁻¹.

Seed yield

Seed yield of pigeonpea increased significantly with the application of compost and under natural farming plots (Fig. 7). The highest seed yield was recorded under compost 12 t ha⁻¹ (13.4 q ha⁻¹), whereas the lowest yield was observed in the control plots (10.1 q ha⁻¹). Although NF plots recorded notable ($p < 0.05$) 11.1% rise in seed yield compared to the control, the yield was still 11.9% and 16.5% lower than those from compost plots of 6 and 12 t ha⁻¹, respectively.

Correlation matrix

The Pearson's correlation matrix was revealed significant relationship among root morphological traits (mean value of all growth stages at 0–15 & 15–30 cm depth) viz., total root length (RL), total root surface area (RSA), total root volume (RV), specific root length (SRL), root growth constant (RGC), root fresh weight (RFW), dry weight (RDW), and nutrient uptake (N, P, K) and grain yield of pigeonpea (Figure S1). There was a highly significant ($r > 0.9$, $p < 0.001$) positive correlation among various traits, RL, RSA, RV, RFW and RDW except SRL at 15–30 cm, which shows a stronger to weaker interpretation. Root morphological traits, including RL, RSA, RV, RFW, RDW and RGC, exhibited highly significant positive correlations with grain yield ($r = 0.81$ to 0.93 , $p < 0.001$), indicating that increased root system development is strongly related to enhanced yield performance. The root traits are most important for nutrient acquisition, particularly for P and K uptake. The correlation between root traits and nutrient uptake: P uptake ($r = 0.91$ – 0.99 , $p < 0.001$) and K uptake ($r = 0.86$ – 0.97 , $p < 0.001$), while moderate to strong correlations were observed with nitrogen (N) uptake ($r = 0.74$ – 0.91 , $p < 0.001$). The specific root length showed uneven and weaker correlation with nutrient acquisition and grain yield. These

results suggest that an enhanced root system facilitates superior nutrient acquisition capacity, particularly for phosphorus and potassium, which consequently translates into improved grain yield.

Prediction model of pigeonpea yield using root traits

The multiple regression equation was developed to determine the contribution of root traits to seed yield. The suggested model equation was:

$$Y = 1.94 - 0.001RL + 0.021RSA - 0.648RV - 0.337RDW + 0.702RFW + 1.635SRL.$$

Where, Y = seed yield (q ha^{-1}), RL = root length (cm), RSA = root surface area (cm^2), RV = root volume (cm^3), RDW = root dry weight (g), RFW = root fresh weight (g), SRL = specific root length (m g^{-1}).

Among the evaluated root traits, RFW, RSA, and SRL showed positive correlations with seed yield, indicated that plants with higher RFW, RSA and SRL value yielded higher seed yield. Conversely, RL, RV and RDW exhibited negative correlations. This model identified RFW as the only significant independent predictor of seed yield among the evaluated root traits ($\beta = 0.702$, $p = 0.011$). The final multiple regression model showed high accuracy ($R^2 = 0.816$, Root mean square error = 0.55). The linear regression analysis curve between seed yield and root fresh weight depicted in Figure S2.

Discussion

Root morphological traits plasticity

Root morphological traits play critical role in determining a crop's capability to access and absorb mineral nutrients from the soil, as well as in deciding the adoption of suitable soil management practices for sustainable crop production. The fertiliser management are important method to regulate the spatial-temporal construction and distribution of the root system³⁷. In the present study, the application of compost (0, 6, 12, 18, and 24 t ha^{-1}), NPK, and natural farming were used to evaluate root morphology traits related to N, P, and K concentrations in the root at different physiological growth stages. The application of compost (12 t ha^{-1}) recorded higher value of root morphological traits (i.e., root length, surface area, and root volume) at different stages in both the soil depths than the conventional farming (CF) and natural farming (NF) plots. For example, 19.0% higher root length was recorded in 12 t ha^{-1} compost applied plots in the 0–15 cm soil than NF plots and 9.0% higher than CF plots, RSA and RV also followed similar trend. These results indicated that the intermediate compost rate (12 t ha^{-1}) was more effective in promoting root proliferation than either lower or higher compost rates.

This response may be measured due to additional supply of nutrients from compost, as a source of organic and mineral nutrients, resulting in an enhancement of root growth, nutrient uptake, biomass accumulation, and ultimately crop yield^{38,39}. The beneficial effects of compost work through two primary mechanisms: direct supplementation of soil fertility with essential and trace elements and enhancement of soil microbial activity. It enhances biological abundance and diversity by providing additional favourable habitat for microorganisms^{40,41}. Therefore, the stimulating impact of compost (12 t ha^{-1}) helps better root penetration, enhances access to nutrients and promotes root elongation and proliferation. However, further increases in the compost (18 and 24 t ha^{-1}) resulted in root growth decline across the sampling stages, possibly due to nutrient imbalances, temporary waterlogging, or reduced soil aeration related to organic amendment. It might be due to compost provide excess amount of readily available nitrogen in nitrate and ammonium forms, which can disturb normal root formation, development and efficient actions^{42,43}.

The root length density (RLD) is directly related to root length, and higher RLD represents an adaptive mechanism that enhances effective root surface area, thereby improving nutrient acquisition efficiency⁴⁴. In the present study, root length density was maximum at flowering stage (90 DAS), particularly under compost treatment (12 t ha^{-1}). The higher RLD indicates greater root occupation per unit soil volume, which enhanced nutrient interception and soil exploration. The advantage of compost 12 t ha^{-1} was evident in both soil layers, especially in the 15–30 cm depth where higher RLD than NF suggested improved access to nutrients beyond the surface horizon. The root volume was enhanced as a consequence of non-stop root branching and increased root length density⁴⁵. In the current study, specific root length did not show a steady and significant response to compost and NF practice across the various sampling stages. The spatial distribution of the root system diverse with the soil layers. The root morphological traits were consistently higher near the plant base in the surface soil layer (0–15 cm) compared to 15–30 cm soil depths. Such non-uniformity of root growth in soil layers, creates heterogeneity in extraction of water and nutrients from the soil profile. Root fresh weight was recorded maximum in 0–15 cm soil layer, which may be attribute to higher soil water and nutrients accessibility in upper layer compared to limited root exploration in deeper soil layers.

Root fresh weight was higher in surface soil layer, allocated to greater soil water and nutrient accessibility in the 0–15 cm layer compared to limited root exploration in deeper soil layers. Similar results have been observed in the present study, where higher root growth was concentrated near the surface layer, confirmed the tendency of pigeonpea roots to allocate greater biomass towards nutrient and moisture-rich surface soil⁴⁶. Furthermore, some studies have reported that large part of root dry weight (80–90% of total RDW) was spread within upper layers (0–20 cm)⁴⁷. The Root morphological traits, including RL, RSA, RV, RFW, RDW, and RLD, revealed similar temporal dynamics; enhanced up to flowering stage (90 DAS) and subsequently declined with crop growth. This was due to higher root system demand in pigeonpea during flowering stage to ensure suitable nutrients uptake for sustaining crop growth and yield formation. The coincidence of maximum root development and peak nutrient uptake at 90 DAS indicated a strong synchronisation between root growth and crop nutrient demand. At flowering, pigeonpea requires substantial nutrient acquisition to support reproductive development, and the larger root systems observed under compost (12 t ha^{-1}) contributed to enhanced nutrient uptake during this critical stage. The reduction in root traits parameters from 90 to 135 DAS suggests progressive root senescence and redistribution of assimilates towards pod and seed development.

The improvement in RLD within the 0–20 cm soil layer at flowering stage with the application of organic amendments⁴⁸. Similarly, in cotton, the application of organic manure (6%) resulted in the highest total root length, total root volume, and root surface area at the bud stage⁴⁹. Although some studies have reported a reduction in root biomass around flowering⁵⁰. The root growth declined from 90 to 135 DAS, which might be due to higher decomposition and turnover of fine roots compared to the coarse roots. However, this finding was primarily related to the finest roots, and our present study lacks the differentiation of root types. A similar pattern of post-flowering root decline has been reported in other studies⁵¹. Moreover, root growth and senescence following flowering are closely linked to reorganisation of photosynthates from vegetative parts to developing pods and seeds⁵². Consequently, an insufficient supply of photosynthates to the root system may induce carbon starvation, leading to a sharp decline in root growth⁵³.

Root morphological traits influencing nutrient acquisition

The significant positive correlations observed between most root morphological traits and N, P and K accumulation under different fertiliser management practices indicate that improved root morphological traits play a critical role in enhancing nutrient concentration in pigeonpea. The widespread root system is characterised by longer root length, high root surface area, larger root volume, and deeper rooting depth, increased nutrient acquisition by enhancing root-soil contact and exploration. In the present study, RL emerged as the most vital trait for phosphorus and potassium uptake. As such, root length and root surface area have been recognised as the representative root traits for P⁵⁴ and K⁵⁵ acquisition. The higher P and K acquisition recorded under compost (12 t ha⁻¹) due to an extensive root system⁵⁶. The comparable relationship was observed in common beans and soybeans⁵⁷.

The higher specific root length in deeper soil layers (15–30 cm) showed more efficiency in nutrient acquisition per unit root biomass investment in the subsoil, where nutrient concentrations were lower. In contrast, the weaker correlation between nitrogen uptake and root traits compared to P and K reflected higher N mobility in soil and the contribution of additional acquisition mechanisms, such as symbiotic nitrogen fixation in pigeonpea⁵⁸. Natural farming plots consistently reported higher nitrogen content across the growth stages; however, this response was related to enhanced biological nitrogen fixation, as nodulation or nitrogen fixation activity was not directly quantified in the present study.

Although natural farming plots consistently exhibited the highest nitrogen content in roots, nitrogen uptake was generally higher under compost (12 t ha⁻¹). This suggests that nutrient uptake influenced not only by nutrient concentration but also by the extent of root system development. The enhanced root length, root surface area, root volume, and root biomass observed under compost application at 12 t ha⁻¹ are mechanistically linked to soil to facilitate greater nitrogen acquisition. The positive correlations between root morphological traits and nutrient uptake further supported that a well-developed root system plays a critical role in determining overall nutrient acquisition in pigeonpea. The synchronisation of nutrient release from organic amendments with peak crop nutrient demand at the flowering stage resulted in maximum nutrient uptake at 90 DAS across all treatments, indicating the importance of temporal matching between nutrient supply and crop requirements for yield production⁵⁹.

Seed yield

Pigeonpea seed yield and root system growth variations are significantly affected by different factors such as farm activities, environmental conditions, types of soil, planting density, water and fertilizer management practices. This study showed that plots with compost (12 t ha⁻¹) achieved higher yields compared to CF and NF plots, and these resulted from higher quantities of root traits in soil depths, as indicated by higher RLD, RSA, root volume, and root biomass. The optimised application of organic amendments prohibited the root growth and significantly increased nutrient uptake and seed yield^{60,61}. The NF plots produced yield significantly similar to CF plots and it is easy, cost-effective and eco-friendly. The yield of compost (12 t ha⁻¹) was 14.6–22.0% higher than compost 18 and 24 t ha⁻¹, respectively and 15.6–19.8% higher than CF and NF plots, respectively. These results suggest that improved root development enhanced nutrient acquisition efficiency, ultimately contributing to greater seed production. Thus, compost (12 t ha⁻¹) should be appropriate for farm and likely be accepted by farmers for higher economic benefits under the conditions of present study.

Relationship between root morphological traits and seed yield under varying nutrient availability

Dissecting and measuring the relationship between root morphological traits and seed yield is crucial for achieving the economic potential of pigeonpea under suitable fertiliser management. The spreading of root system in various soil depths primarily determined the absorption of nutrients from soil and increased grain yield³⁷. In comparison with inorganic fertilisers, the compost (12 t ha⁻¹) application significantly increased root growth in surface and subsurface layers, and hence the well-developed root system further enhanced nutrient use from various layers. Inorganic fertilizer application significantly decreased root system growth in surface and subsurface depths, weakened nutrient acquisition, and did not fulfil the requirements for grain yield formation. Some studies reported that optimized fertilizer applications increased root growth and reduced root dry weight in 0–60 cm depth, although significantly increased the RDW in the 60–120 cm soil depths and total root dry biomass in all depths, and confirmed the higher grain yield⁴⁴. Smaller root length density in subsoil layer reduces crop water and nutrient utilisation; however, higher RLD in the soil layers might capture water and nutrients more efficiently, and economic yields could be highly increased⁶². Our study had comparable findings to these studies. Compared to NPK, NF and various compost rates (6, 18 & 24 t ha⁻¹) achieved more yields and higher quantities of root systems in surface and subsurface depths, as indicated by higher RL, RLD, RSA, RV, RFW and RDW and correlation coefficients between grain yield and root traits. These suggested that rich root system in

different soil depths could show important roles in enhancing grain yields. Therefore, to encourage the growth of root morphological traits, quantity and quality in various soil depths may be considered a major technology to improve economic grain yield. The positive correlations between seed yield and RFW, RSA and SRL further support the importance of root system development in determining crop productivity.

Relationship between temporal nutrient dynamics and seed yield

The nutrient content of root was not constant; however, measurement of root N, P and K at different physiological stages provided detailed information on root nutrient dynamics and nutrient requirements for achieving higher grain yield. As previously reported, grain yield was strongly related to phosphorus content during the flowering stage⁶³. In our present study, linear regression analysis was used to analyse the relationship between grain yield, and N, P and K content at different growth stages (Fig. 8A, B & C). Phosphorus exhibited most reliable association with grain yield across all growth stages (R^2 : 0.78–0.85). The vegetative growth (45 DAS) showed highly significant linear relationship ($y = 37.27x + 6.35$, $R^2 = 0.85$), indicating that early P nutrition establishes the basis for yield potential by supporting root system and early strength. The relationship at flowering stage was ($y = 14.35x + 8.06$, $R^2 = 0.83$) and pod filling stage ($y = 83.75x + 8.22$, $R^2 = 0.78$), representing the critical role of phosphorus in seed formation. The slope regression coefficient reported highest at 45 DAS (37.27) and 135 DAS (83.75), showing that phosphorus availability during these stages has an unpredictable impact on grain yield. During these stages, limited availability of phosphorus severely decreased grain production⁶⁴. These findings are also reported in maize and wheat, where enhanced phosphorous availability significantly improved root traits and elevated the root contribution to total plant dry weight (up to 24.07%) and enhanced grain yield⁶⁵.

Similarly, potassium recognized the relationship with grain yield at flowering stage (90 DAS: $y = 5.84x + 7.99$, $R^2 = 0.93$), representing 93% yield variation with K supply. It might be due to critical growth stage (90DAS) required higher stomatal conductance for optimal photosynthesis, activation of enzymes involved in protein synthesis, and enabling photosynthates translocation from source to sink⁶⁶. The growth stages at 45 DAS and 135 DAS showed ($R^2 = 0.75$), indicating that K content throughout the growth stages contributed to sustainable grain production; however flowering stage denoted most K-sensitive stage. Nitrogen showed correlation with grain yield ($R^2 = 0.32$ – 0.54) compared to phosphorous and potassium content, with highest ($R^2 = 0.54$) at flowering stage. The stronger relationships observed for phosphorus and potassium than for nitrogen suggest that P and K availability were more limiting to grain production under the present conditions. While nitrogen is partly supplied through biological nitrogen fixation in pigeonpea, phosphorus and potassium acquisition depend more directly on root exploration and nutrient availability in soil. Hence, root morphological traits exhibited stronger correlations with P and K uptake than with nitrogen uptake. The nutrient acquisition efficiency attained through optimized root morphological traits, leading to higher grain yield. Therefore, encouraging crop and fertilizer management strategies that specifically focus on root system architecture and stage-specific nutrient requirement, especially phosphorous and potassium could be a vital technique to enhance resource use efficiency and yield stability.

Conclusions

This study illustrates that root development is an important mediator through which soil management practices affect nutrient uptake and yield of pigeonpea. Root phosphorus is the strongest contributor to seed yield, followed by potassium and nitrogen, highlighting the importance of root-mediated nutrient acquisition in yield formation. Compost application at 12 t ha^{-1} facilitate the favourable conditions to promote the effective root system development and could be an eco-friendly alternative for maintaining pigeonpea yield. Finding of the study expands our understanding of spatio-temporal regulation of root system and 90 days after sowing of pigeonpea is ideal time for root study that governing nutrient uptake and crop productivity most. The study further illustrated traits related to the roots can be used as functional indicators to determine the crop response to nutrient management and mechanism for yield difference. Hence, integration of root functional traits into nutrient management could be an innovative tool in improving resource-use efficiency, crop yield and resilience of pigeonpea and other crops in sustainable agriculture.

Future research should investigate the long-term effects of compost-based nutrient management on root system architecture, integrating root phenotyping and multi-location field evaluations would improve understanding of root-mediated nutrient acquisition and support the development of resilient and sustainable pigeonpea production systems.

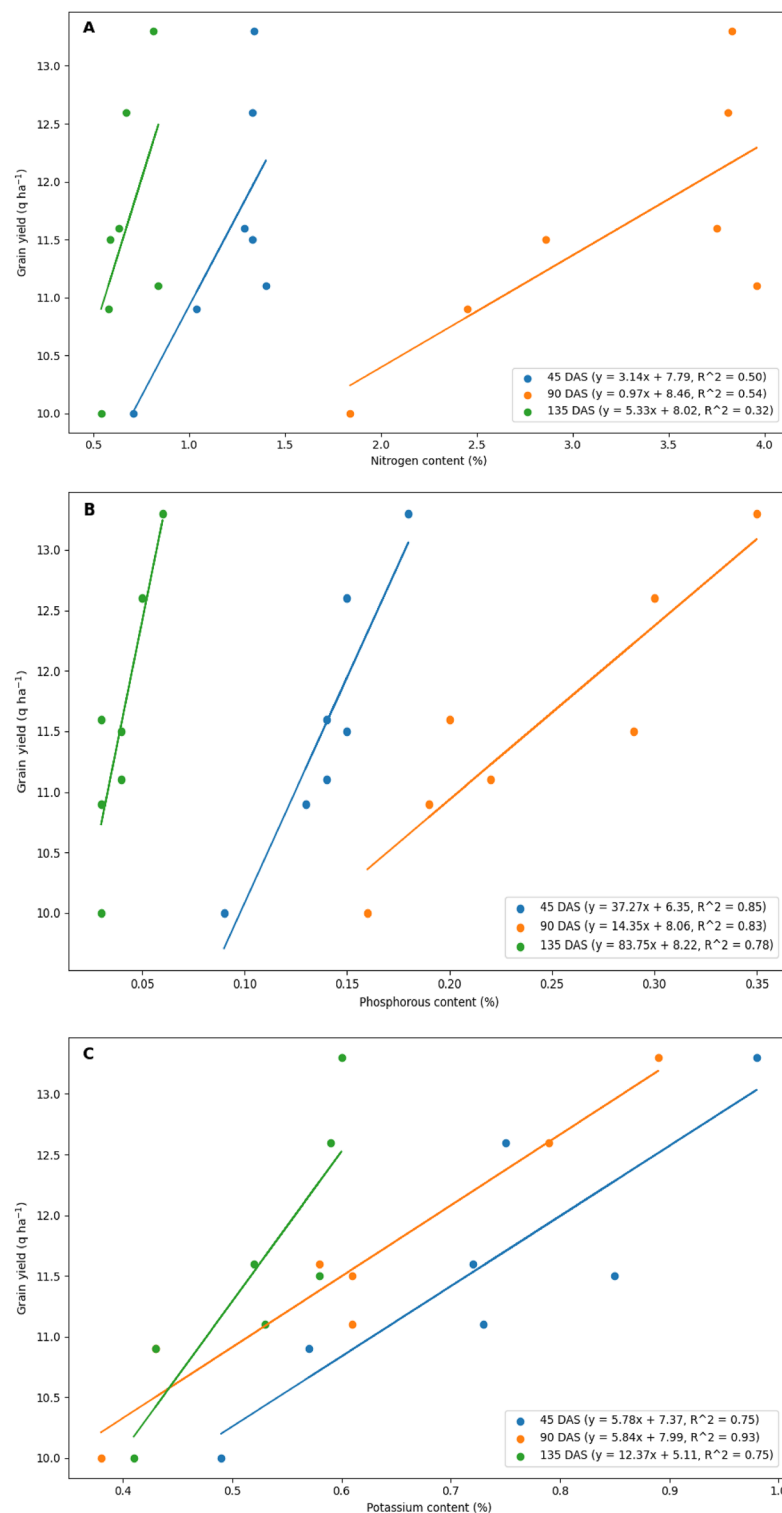


Fig. 8. Response of root nitrogen (A), phosphorous (B) and potassium (C) content at 45, 90 and 135 days after sowing in 0–30 cm soil depth to grain yield of pigeonpea under different crop management strategies.

Data availability

The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

Received: 23 March 2026; Accepted: 2 July 2026

Published online: 07 September 2026

References

1. Locali-Pereira, A. R. et al. Pigeon pea, an emerging source of plant-based proteins. *ACS Food Sci. Technol.* **3**, 1777–1799. <https://doi.org/10.1021/acsfoodscitech.3c00268> (2023).
2. Gerrano, A. S., Moalafi, A., Seepe, H. A., Amoo, S. & Shimelis, H. Nutritional and phytochemical compositions and their interrelationship in succulent pods of pigeonpea (*Cajanus cajan* [L.] Millsp.). *Heliyon* **8**, e09078. <https://doi.org/10.1016/j.heliyon.2022.e09078> (2022).
3. Food and Agriculture Organization of the United Nations. *FAOSTAT Statistical Database* (FAO, 2022).
4. Fatokimi, E. O. & Tanimonure, V. A. Analysis of the current situation and future outlooks for pigeon pea (*Cajanus cajan*) production in Oyo State, Nigeria: A Markov Chain model approach. *J. Agric. Food Res.* **6**, 100218. <https://doi.org/10.1016/j.jafr.2021.100218> (2021).
5. Singh Gahoonia, T. & Nielsen, N. E. Root traits as tools for creating phosphorus efficient crop varieties. *Plant. Soil.* **260**, 47–57 (2004).
6. Kan, Z. R. et al. Responses of grain yield and water use efficiency of winter wheat to tillage in the North China Plain. *Field Crops Res.* **249**, 107760. <https://doi.org/10.1016/j.fcr.2020.107760> (2020).
7. Zhou, H. et al. The interaction between wheat roots and soil pores in structured field soil. *J. Exp. Bot.* **72**, 747–756. <https://doi.org/10.1093/jxb/eraa475> (2021).
8. Giller, K. E. et al. The future of farming: Who will produce our food. *IFAD Res. Ser.* **83**, 1073–1099. <https://doi.org/10.1007/s12571-021-01184-6> (2022).
9. Chowdhuri, A., Chaudhary, M., Purakayastha, T. J., Singh, T. & Sinha, K. G. R. Optimizing soil fertility and climate resilience: Superiority of organic farming in enhancing carbon sequestration and nitrogen supply. *J. Environ. Manage.* **393**, 127131. <https://doi.org/10.1016/j.jenvman.2025.127131> (2025).
10. Hodkashia, S. et al. Effect of FYM and inorganic fertilizers on growth, yield of wheat (*Triticum aestivum* L.) and soil properties of Red Alfisol. *Asian J. Soil. Sci. Plant. Nutr.* **10**, 403–413 (2024).
11. Srivastav, A. L. et al. Sustainable options for fertilizer management in agriculture to prevent water contamination: A review. *Environ. Dev. Sustain.* **26**, 8303–8327. <https://doi.org/10.1007/s10668-023-03117-z> (2024).
12. Singh, V. K., Malhi, G. S., Kaur, M., Singh, G. & Jatav, H. S. Use of organic soil amendments for improving soil ecosystem health and crop productivity. *Ecosyst. Serv.* **12**, 45 (2022).
13. Chen, L., de Haro Marti, M., Moore, A. & Falen, C. The composting process. *Dairy. Manure Compost Prod. Use Ida.* **2**, 513–532 (2011).
14. Suganthi, M., Renuka Rani, B. & Sharma, N. R. Natural farming for sustainable agriculture. *Nammazhvar Organic Farming Research Centre, Tamil Nadu Agricultural University, Coimbatore & National Institute of Agricultural Extension Management (MANAGE), Hyderabad*, 122–138 (2023).
15. Goulding, K., Jarvis, S. & Whitmore, A. Optimizing nutrient management for farm systems. *Philosophical Trans. Royal Soc. B: Biol. Sci.* **363**, 667–680. <https://doi.org/10.1098/rstb.2007.2177> (2008).
16. Dhaliwal, S. S. et al. Enhanced organic carbon triggers transformations of macronutrients, micronutrients, and secondary plant nutrients and their dynamics in the soil under different cropping systems: A review. *J. Soil. Sci. Plant. Nutr.* **24**, 5272–5292. <https://doi.org/10.1007/s42729-024-01907-6> (2024).
17. Varatharajan, T. et al. Effect of integrated crop management practices on root-shoot characteristics, yield and nutrient harvest index of pigeon pea (*Cajanus cajan*) under irrigated North Indian Plains. *Ind. J. Agron.* **64**, 270–274 (2019).
18. Bansal, R. & Srivastava, J. P. Effect of waterlogging on root anatomy and nitrogen distribution in pigeonpea (*Cajanus cajan* (L.) Millsp.). *Indian J. Plant. Physiol.* **22**, 130–134. <https://doi.org/10.1007/s40502-016-0247-y> (2017).
19. Kellermeier, F. et al. Analysis of the root system architecture of *Arabidopsis* provides a quantitative readout of crosstalk between nutritional signals. *Plant. Cell.* **26**, 1480–1496. <https://doi.org/10.1105/tpc.113.122101> (2014).
20. Rautela, S., Bains, G., Singh, D. K., Jahan, S. & Thi, T. Effect of organic, inorganic and integrated nutrient amendments on growth parameters of basmati rice (*Oryza sativa* L.). *Agric. Ecosyst. Environ.* **140**, 339–353 (2022).
21. Chinthapalli, B., Dibar, D. T., Chitra, D. V. & Leta, M. B. A comparative study on the effect of organic and inorganic fertilizers on agronomic performance of faba bean (*Vicia faba* L.) and pea (*Pisum sativum* L.). *Agric. Forestry Fisheries.* **4**, 263–268. <https://doi.org/10.11648/j.aff.20150406.15> (2015).
22. Wang, Z. et al. From phenotyping to genetic mapping: Identifying water-stress adaptations in legume root traits. *BMC Plant Biol.* **24**, 749. <https://doi.org/10.1186/s12870-024-05477-8> (2024).
23. Gaiotti, F. et al. Influence of compost addition on soil properties, root growth and vine performances of *Vitis vinifera* cv. Cabernet Sauvignon. *Sci. Hort.* **225**, 88–95. <https://doi.org/10.1016/j.scienta.2017.06.052> (2017).
24. Fageria, N. K., Gheyi, H. R., Carvalho, M. C. S. & Moreira, A. Root growth, nutrient uptake and use efficiency by roots of tropical legume cover crops as influenced by phosphorus fertilization. *J. Plant Nutr.* **39**, 781–792. <https://doi.org/10.1080/01904167.2015.1088020> (2016).
25. Iqbal, A. et al. Organic manure coupled with inorganic fertilizer: An approach for the sustainable production of rice by improving soil properties and nitrogen use efficiency. *Agronomy* **9**, 651. <https://doi.org/10.3390/agronomy9100651> (2019).
26. Chen, X. et al. Root physiological adaptations that enhance the grain yield and nutrient use efficiency of maize (*Zea mays* L.) and their dependency on phosphorus placement depth. *Field Crops Res.* **276**, 108378. <https://doi.org/10.1016/j.fcr.2021.108378> (2022).
27. Böhm, W. *Methods of Studying Root Systems* (Springer Science & Business Media, 2012).
28. Costa, C. et al. A sampling method for measurement of large root systems with scanner-based image analysis. *Agron. J.* **92**, 621–627. <https://doi.org/10.2134/agronj2000.924621x> (2000).
29. Myers, D. B., Kitchen, N. R., Sudduth, K. A., Sharp, R. E. & Miles, R. J. Soybean root distribution related to claypan soil properties and apparent soil electrical conductivity. *Crop Sci.* **47**, 1498–1509. <https://doi.org/10.2135/cropsci2006.07.0460> (2007).
30. Föhse, D., Claassen, N. & Jungk, A. Phosphorus efficiency of plants: II. Significance of root radius, root hairs and cation-anion balance for phosphorus influx in seven plant species. *Plant. Soil.* **132**, 261–272. <https://doi.org/10.1007/BF00010407> (1991).
31. Ostonen, I. et al. Specific root length as an indicator of environmental change. *Plant. Biosystems.* **141**, 426–442. <https://doi.org/10.1080/11263500701626069> (2007).
32. Burns, I. G. & Hutsby, W. Critical comparison of the vanadomolybdate and the molybdenum blue methods for the analysis of phosphate in plant sap. *Commun. Soil Sci. Plant Anal.* **17**, 839–852. <https://doi.org/10.1080/00103628609367756> (1986).
33. Jackson, M. L. *Soil Chemical Analysis-Advanced Course: A Manual of Methods Useful for Instruction and Research in Soil Chemistry, Physical Chemistry of Soils, Soil Fertility, and Soil Genesis* (1973).
34. Matejovic, I. Total nitrogen in plant material determined by means of dry combustion: A possible alternative to determination by Kjeldahl digestion. *Commun. Soil Sci. Plant Anal.* **26**, 2217–2229. <https://doi.org/10.1080/00103629509369441> (1995).
35. Sharma, V., Singh, M. J. & Khokhar, A. K. Productivity, nutrient uptake and soil properties as influenced by integrated nutrient management in maize-wheat cropping system under rainfed conditions of sub-montane Punjab. *Agricultural Res. J.* **57**, 839–847. <https://doi.org/10.5958/2395-146X.2020.00123.4> (2020).
36. Gomez, K. A. & Gomez, A. A. *Statistical Procedures for Agricultural Research*. John Wiley & Sons (1984).
37. Wang, Y. et al. Effects of cultivation patterns on winter wheat root growth parameters and grain yield. *Field Crops Res.* **156**, 208–218. <https://doi.org/10.1016/j.fcr.2013.11.017> (2014).
38. Dwivedi, S. et al. Soil-test-crop-response based nutrient scheduling can improve soybean-wheat productivity and system sustainability. *BMC Plant Biol.* **24**, 1203. <https://doi.org/10.1186/s12870-024-05890-z> (2024).

39. Hou, Y. H., Kuo, C. Y., Lai, T. Y. E., Xiao, M. H. & Ho, S. L. Transcriptional corepressor OsTPR1 regulates tillering and lateral root development in rice. *Sci. Rep.* **15**, 26430. <https://doi.org/10.1038/s41598-025-10224-6> (2025).
40. Saleem, M., Law, A. D., Sahib, M. R., Pervaiz, Z. H. & Zhang, Q. Impact of root system architecture on rhizosphere and root microbiome. *Rhizosphere* **6**, 47–51. <https://doi.org/10.1016/j.rhisph.2018.02.003> (2018).
41. Wang, Y. et al. The diversity of microbial community and function varied in response to different agricultural residues composting. *Sci. Total Environ.* **715**, 136983. <https://doi.org/10.1016/j.scitotenv.2020.136983> (2020).
42. Zhang, J. et al. *Phlomis rotata* adapts to low-nitrogen environments by promoting root growth and increasing root organic acid exudate. *BMC Plant Biol.* **24**, 1234. <https://doi.org/10.1186/s12870-024-05962-0> (2024).
43. Kaur, D. & Singh, G. Nutrient uptake, nutrient use efficiency and residual soil nutrient status in kabuli chickpea (*Cicer arietinum* L.) as influenced by microbial inoculation and phosphorus application. *Commun. Soil Sci. Plant Anal.* **56**, 2036–2051. <https://doi.org/10.1080/00103624.2025.2489116> (2025).
44. Chen, J. et al. Nitrogen fertilization increases root growth and coordinates the root-shoot relationship in cotton. *Front. Plant Sci.* **11**, 880. <https://doi.org/10.3389/fpls.2020.00880> (2020).
45. Lynch, J. P. & Brown, K. M. New roots for agriculture: Exploiting the root phenome. *Philosophical Trans. Royal Soc. B: Biol. Sci.* **367**, 1598–1604. <https://doi.org/10.1098/rstb.2011.0243> (2012).
46. Sheldrake, A. R. & Narayanan, A. Growth, development and nutrient uptake in pigeonpeas (*Cajanus cajan*). *J. Agric. Sci.* **92**, 513–526. <https://doi.org/10.1017/S0021859600053752> (1979).
47. Dwyer, L. M. et al. Root mass distribution under conventional and conservation tillage. *Can. J. Soil. Sci.* **76**, 23–28. <https://doi.org/10.4141/cjss96-004> (1996).
48. Zhao, L. et al. Organic amendments improve wheat root growth and yield through regulating soil properties. *Agron. J.* **111**, 482–495. <https://doi.org/10.2134/agronj2018.04.0247> (2019).
49. Zhang, Z., Dong, X., Wang, S. & Pu, X. Benefits of organic manure combined with biochar amendments to cotton root growth and yield under continuous cropping systems in Xinjiang, China. *Sci. Rep.* **10**, 4718. <https://doi.org/10.1038/s41598-020-61118-8> (2020).
50. Salter, P. J. & Drew, D. H. Root growth as a factor in the response of *Pisum sativum* L. to irrigation. *Nature* **206**, 1063–1064. <https://doi.org/10.1038/2061063b0> (1965).
51. Williams, J. L. et al. Relationships between roots, the stay-green phenotype, and agronomic performance in barley and wheat grown in semi-arid conditions. *Plant. Phenome J.* **5**, 220050. <https://doi.org/10.1002/ppj2.20050> (2022).
52. Marcel, T. Senescence in annual plants: A plant to survive. *Trends Plant. Sci.* **12**, 548–558 (2007).
53. Vance, C. P., Johnson, L. E. B., Halvorsen, A. M., Heichel, G. H. & Barnes, D. K. Histological and ultrastructural observations of *Medicago sativa* root nodule senescence after foliage removal. *Can. J. Bot.* **58**, 295–309. <https://doi.org/10.1139/b80-030> (1980).
54. Gahoonia, T. S., Ali, R., Malhotra, R. S., Jahoor, A. & Rahman, M. M. Variation in root morphological and physiological traits and nutrient uptake of chickpea genotypes. *J. Plant. Nutr.* **30**, 829–841. <https://doi.org/10.1080/15226510701373213> (2007).
55. Høgh-Jensen, H. & Pedersen, M. B. Morphological plasticity by crop plants and their potassium use efficiency. *J. Plant. Nutr.* **26**, 969–984. <https://doi.org/10.1081/PLN-120020069> (2003).
56. Paez-Garcia, A. et al. Root traits and phenotyping strategies for plant improvement. *Plants* **4**, 334–355. <https://doi.org/10.3390/plants4020334> (2015).
57. Ao, J., Fu, J., Tian, J., Yan, X. & Liao, H. Genetic variability for root morph-architecture traits and root growth dynamics as related to phosphorus efficiency in soybean. *Funct. Plant. Biol.* **37**, 304–312. <https://doi.org/10.1071/FP09215> (2010).
58. Giller, K. E. *Nitrogen Fixation in Tropical Cropping Systems* (CABI, 2001).
59. Palm, C., Blanco-Canqui, H., DeClerck, F., Gater, L. & Grace, P. Conservation agriculture and ecosystem services: An overview. *Agric. Ecosyst. Environ.* **187**, 87–105. <https://doi.org/10.1016/j.agee.2013.10.010> (2014).
60. Griffiths, M. et al. Optimisation of root traits to provide enhanced ecosystem services in agricultural systems: A focus on cover crops. *Plant. Cell. Environ.* **45**, 751–770. <https://doi.org/10.1111/pce.14247> (2022).
61. Pindke, M. S. et al. Impact of tillage and fertilizer management on soybean-cotton rotation system: effects on yield, plant nutrient uptake, and soil fertility for sustainable agriculture. *Sci. Rep.* **15**, 9991. <https://doi.org/10.1038/s41598-025-95116-5> (2025).
62. Lecompte, F., Ozier-Lafontaine, H. & Pagès, L. The relationships between static and dynamic variables in the description of root growth: Consequences for field interpretation of rooting variability. *Plant. Soil* **236**, 19–31. <https://doi.org/10.1023/A:1011924529885> (2001).
63. Mahanta, D. et al. Influence of phosphorus and biofertilizers on soybean and wheat root growth and properties. *Field Crops Res.* **166**, 1–9. <https://doi.org/10.1016/j.fcr.2014.06.016> (2014).
64. Vance, C. P., Uhde-Stone, C. & Allan, D. L. Phosphorus acquisition and use: critical adaptations by plants for securing a nonrenewable resource. *New Phytol.* **157**, 423–447. <https://doi.org/10.1046/j.1469-8137.2003.00695.x> (2003).
65. Harish, M. N. et al. P. N. Double zero-tilled bed-planting and optimized P-fertilization in maize-wheat rotation: An empirical investigation on root architecture, carbon-phosphorus dynamics, and soil carbon pools. *J. Environ. Manage.* **392**, 126668. <https://doi.org/10.1016/j.jenvman.2025.126668> (2025).
66. Pettigrew, W. T. Potassium influences on yield and quality production for maize, wheat, soybean and cotton. *Physiol. Plant.* **133**, 670–681. <https://doi.org/10.1111/j.1399-3054.2008.01073.x> (2008).

Acknowledgements

The authors are extremely grateful to the ICAR-Indian Agricultural Research Institute, New Delhi, for providing all the necessary infrastructure and facilities for carrying out this research. Authors are also thankful to Manipal University Jaipur, India, for providing financial support from university grant.

Author contributions

Seema Hodkashia: Writing – review & editing, investigation, data analysis, original draft preparation; Sarvendra Kumar: Writing – review & editing, validation, supervision, conceptualization, data curation, manuscript formatting and editing; Dibakar Mahanta: field experimentation Tapan Jyoti Purakayastha: conceptualization; Rakesh Pandey: root analysis; Debashis Mandal: conceptualization; Rosin K.G.: Review and editing of manuscript; Anil Kumar Mishra: conceptualization; Sangeeta Paul: conceptualization and manuscript editing; V. Mubashir Sadiq, Sumitra Kumawat, Arijit Chowdhuri: root analysis, data analysis. Arun Kumar: Review and editing of manuscript, fund acquisition.

Funding

Open access funding provided by Manipal University Jaipur. Ms. Seema Hodkashia received financial support from ICAR- Indian Agricultural Research Institute, New Delhi, in the form of an IARI fellowship. The authors express their sincere gratitude to the Director, ICAR-Indian Agricultural Research Institute, New Delhi, for

supporting the successful conduct of the research work. This research received no external funding.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-61158-6>.

Correspondence and requests for materials should be addressed to S.K. or A.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026